

Migratory birds in urban landscapes

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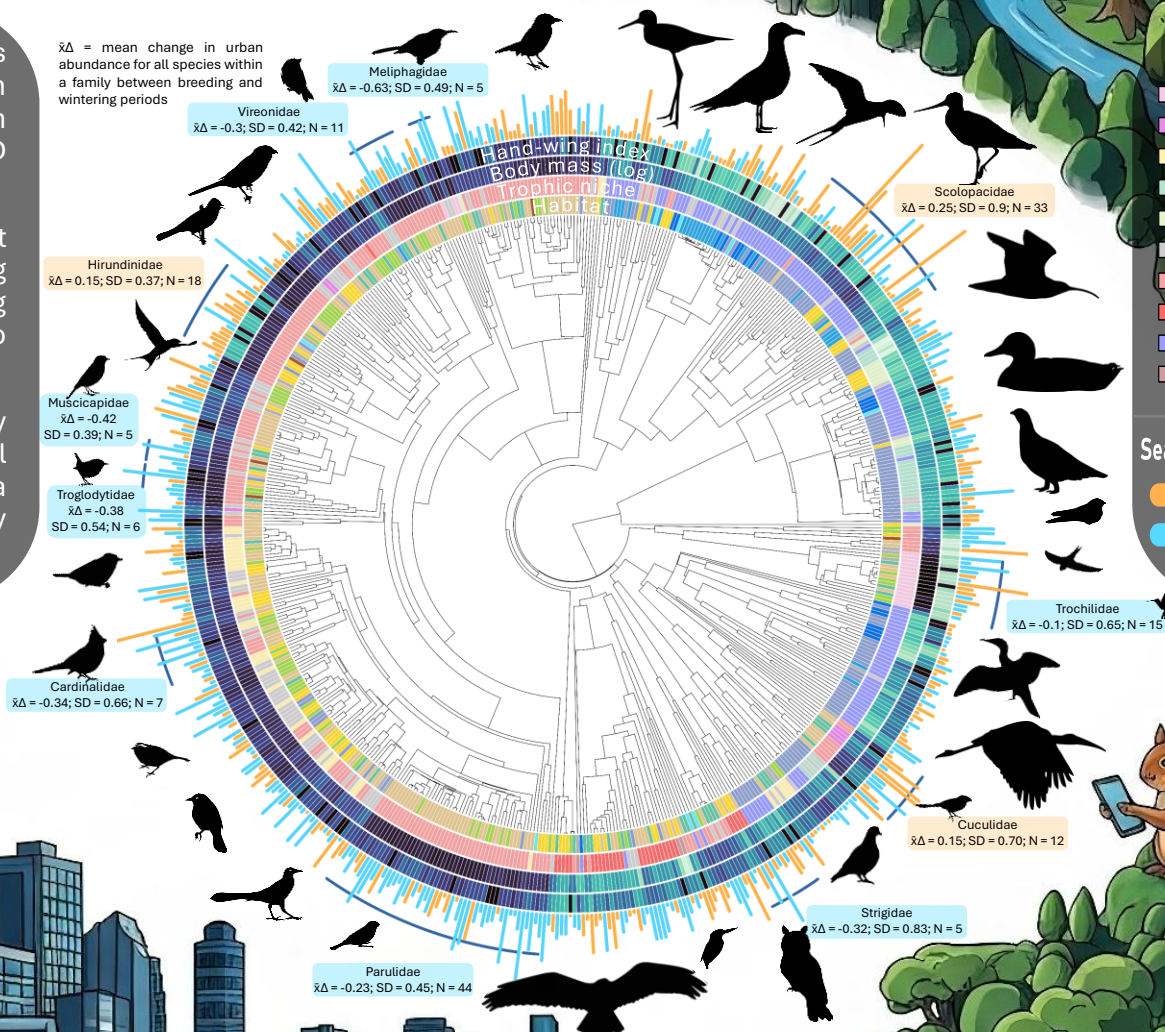
Comparing seasonal abundancies in human-altered areas

- Migratory birds track seasonal changes in resources, but these seasonal patterns likely differ between urban and natural landscapes.
- Urban areas can offer breeding sites and may buffer against harsh non-breeding conditions.
- Yet, the effects of human-altered landscapes on seasonal distribution of migratory birds remain poorly understood.
- Here, we map seasonal changes in abundances of migratory bird species, identify key life-history predictors and quantify consequences for population dynamics.

Results

- Across species, migratory birds seem to be more abundant in urban areas during wintering than during breeding (mean -0.05 , SD 0.51).
- 197 species were more abundant in urban areas during breeding ($\Delta > 0.1$), 243 during wintering ($\Delta < -0.1$), and 128 showed no significant change ($-0.1 \leq \Delta \leq 0.1$).
- Species from the same family often showed similar seasonal shifts in abundance, suggesting a role of shared evolutionary history (detailed analysis in preparation).

$\bar{\Delta}$ = mean change in urban abundance for all species within a family between breeding and wintering periods



Trophic Niche

- Nectarivore
- Frugivore
- Granivore
- Herbivore aquatic
- Herbivore terrestrial
- Omnivore
- Invertivore
- Vertivore
- Aquatic predator
- Scavenger

Habitat

- Grassland
- Shrubland
- Woodland
- Forest
- Wetland
- Riverine
- Coastal
- Marine
- Rock
- Human Modified

Seasonal changes in abundancies in urban areas

- More abundant during breeding
- More abundant during wintering

Future steps

1. A formal analysis of relationships between seasonal changes in abundances and species' life-history traits.
2. Compile data on long-term population trends and quantify relationships between seasonal changes in abundances and population dynamics.